Solutions

1. This question addresses transformers [25]

a) Some written questions on transformers

i) What is it that a transformer does that is highly beneficial in a power system? [1]

It can increase/decrease the voltage/current [1]

ii) Why is it beneficial? [1]

In a case where we want to transfer energy over long distances increasing the voltage reduces the current and therefore reducing the current squared losses. [1]

iii) How does it do it? [1]

It takes advantage of Faraday law – rate of change of flux linkage is the induced emf – two winding on a common core will share the flux will see the same flux changing but will have different flux linkages because of different number of turns. So, the emf (voltages) will be different either above, equal or below. [1]

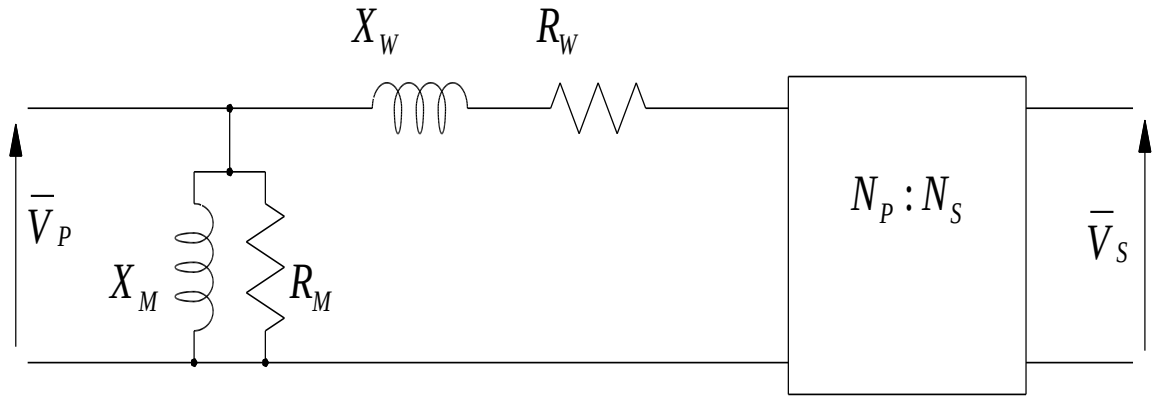
iv) Most transformers in the world today are step down – why is this? [1]

Because at the generators large amounts of energy are involved and these use step-up transformers to transport the energy to the consumer. Nearer the consumer you have to step it down and the energy usage is smaller meaning more step-down transformers [1]

b) A 50 Hz 100 kVA 2.4 kV to 240 V single phase step-down transformer has the following parameters for its approximate equivalent circuit

Winding resistance	$R_W = 0.8\ \Omega$
Winding reactance	$X_W = 5.0\ \Omega$
Magnetizing resistance	$R_M = 7\ \text{k}\Omega$
Magnetising reactance	$X_M = 1\ \text{k}\Omega$

This FIGURE is part of the solution not necessary but would be good for them to draw it



i) Calculate the turns ratio [1]

$N_P/N_S = 10$ [1]

ii) Calculate the rated current on the primary side [1]

$I_{P,rated} = 100/2.4 = 41.66$ A [1]

iii) Calculate the rated current on the secondary side [1]

$I_{S,rated} = 100/0.24 = 416.66$ A [1]

c) For the transformer in (b), rated voltage is applied at the primary side and the secondary side is open circuit. Answer the questions below.

i) Calculate the voltage on the secondary side [1]

V_s = 240 V – this is the approximate equivalent circuit so on O/C the ideal transformer sees the primary voltage with no voltage drop across wind impedances – not correct but it is the approximate equivalent circuit [1]

ii) Calculate the winding losses [1]

Zero again approx. equivalent circuit [1]

iii) Calculate the magnetizing losses [1]

$= (V_P)^2/R_M = 822.86$ Watts [1]

iv) Calculate the power factor stating whether it is lagging or leading [2]

$I_P = (2400/7000) - j\,(2400/1000) = 0.34 - 2.40i$ $\varphi(I_p) = -81.87$ degrees [1]

Power factor = Cos (-81.87) = 0.142 lagging [1]

v) Calculate the efficiency [1]

Zero as it is open circuit, so no energy delivered to the load [1]

d) For the transformer in (b), rated voltage is applied at the primary side and a load of 0.5 + j 0.1 Ω is connected to the secondary side. Answer the questions below.

i) Calculate the voltage on the secondary side [3]

Z_{load} = 0.5 + j 0.1

Z_{load} on primary side = 50 + j 10

Z_{load} on primary side added to Z_w in series = 50.8 + j15 Ω

Secondary voltage V_s = (1/10) 2400 $\frac{50+j\ 10}{50.8+j15}$ = 231.04<-5.14° = 230.1– j 20.70 V [3]

ii) Calculate the current on the secondary side [3]

Is = (231.04<-5.14°)/(0.5 + j0.1) = 453.1 <-16.45 ° =434.55– j 128.31 A

Over its rated current of 416.66 – the very bright students may notice this. [3]

iii) Calculate the real and reactive power of the load [3]

S = (230.1– j 20.70)*(434.55+j 128.31) [1]

P = 102.65 kW [1]

Q = 205.30 kVar [1]

iv) Calculate the efficiency [3]

Magnetizing losses = $(2400)^2/7000$ = 822.9 W [1]

Winding losses = $I_p^2 * R_w$=$(453.1/10)^2$*(0.8) = 1642.4 W [1]

Power in load = 102.65 kW

Efficiency = 100%* (102.65 k)/(102.65 k +1642.4+822) = 97.65 % [1]

2. This question addresses 3 phase, synchronous generators and inverters [20]

a) A 3-phase 500KVA synchronous generator is star connected to an infinite bus at 6kV.

The speed of rotation is 1200 revolutions per minute.

i) What is the frequency at the infinite bus [1]

60 Hz because for 50 Hertz this would have a non-integer number of pole pairs (3000/1200=2.5). [1]

ii) How many pole pairs does the generator have [1]

3 because 60 Hz is 3600 rpm and 3600/3 = 1200 [1]

iii) If the generator is generating 450 KW calculate the Torque [2]

T = P/Speed = 450000/(1200/60*2π) = 3581.0 NM [2]

iv) If the torque increases by 5 % calculate the mechanical speed of the machine and the power being generated. [2]

Speed does not change it is a synchronous generator – power will increase by 5 % = 472.5 kW [2]

v) For this increased torque the power factor is unity calculate the current [2]

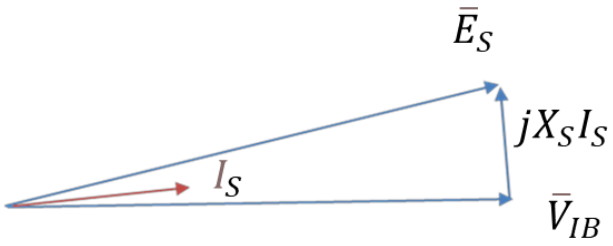
$V_s = 6000/\sqrt{3} = 3464.1V$
$I_s = (472.5x10^3)/(\sqrt{3}x6000) = 45.46\ A$

[2]

vi) If the load angle $\delta$is 20° and generator emf is 3.5kV ignoring armature resistance calculate the synchronous reactance and draw the phasor diagram for this generator.
[4]

$X_s = (3500*3464.1)*\sin(20) / (472.5x10^3)/3) = 26.35\ \Omega$/per phase [2]

Phasor Diagram



[2]

b) A 3-phase permanent magnet synchronous motor (PMSM) is driven by a three-phase inverter.

i) At 3000RPM, the modulator frequency of the inverter is 50Hz. What is the modulator frequency if the speed of the PMSM is changed to 600RPM? [4]

ii) The inverter has dc input voltage (rail-to-rail) of 1000V, which is equivalent to $V_{DC+}=V_{DC-}=500V$. Write a sinusoidal PWM modulator function to generate a 30Hz inverter voltage with an amplitude of 300V per phase (you only need to write the modulator for one of the phases).[2]

iii) At 1000RPM, the EMF of the motor is 100V (line-to-line RMS value). The dc input voltage (rail-to-rail) of the inverter is 1000V. What is the modulation ratio of the inverter at a speed of 3000PRM? [2]

3. This question addresses power flow in lines and cables. [25]

a) Consider the characteristics of single-phase overhead lines.

i) An overhead line is found to have an inductance per unit length of 100 μH/km and is operated at 50 Hz and 290kV. For a line with a length of 150km, calculate the maximum power transferred along the line. [5]

ii) For the line in i), if the maximum load angle is limited to 15°, calculate the maximum current. [5]

iii) An overhead line has a series impedance of 0.65 + j 0.35 Ω. The voltage at the sending end is 15 kV. A real power of +11.0 MW is measured flowing into the line at the sending end. The voltage at the receiving end is measured at 14.3kV. Estimate the reactive power as seen at the sending end. [5]

Solutions:

i)

Use standard equation $P = \frac{V_R V_S}{X_{SR}} \sin(\delta)$

$X_{SR} = 2\pi\, f\, L'l = 2\pi \times 50 \times 100 \times 10^{-6} \times 150 = 4.712\Omega$

The maximum power is achieved when the power angle equals 90 degrees, which is calculated as

$$P_{max} = \frac{V_R V_S}{X_{SR}} \sin(90°) = \frac{(290 \times 10^3)^2}{4.712} \times 1 = 17.847 \text{ GW}$$

ii)

Let the magnitude of the voltage phasor difference between the sending and receiving end be ΔV. The maximum voltage difference magnitude can be calculated from the cos law

$V_R^2 + V_S^2 - 2V_R V_S \cos\delta_{max} = \Delta V_{max}^2$, and the maximum voltage difference is

$$\Delta V_{max} = \sqrt{2 \times 290^2 \times 10^6 - 2 \times 290^2 \times 10^6 \times \cos 15°} = 75.71kV$$

The voltage difference between the sending and receiving end is caused by the line impedance, thus we have $I_{max} = \frac{\Delta V_{max}}{X_{SR}} = \frac{75.71 \times 10^3}{4.712} = 16.07\ kA$.

iii)

Use standard equation $\Delta V = |V_S| - |V_R| \approx \frac{R\,P_S + X\,Q_S}{|V_S|}$

So $Q_S = \frac{\Delta V |V_S| - R P_S}{X} = \frac{0.7 \times 10^3 \times 15 \times 10^3 - 0.65 \times 11.0 \times 10^6}{0.35} = 9.571\text{MVar}$

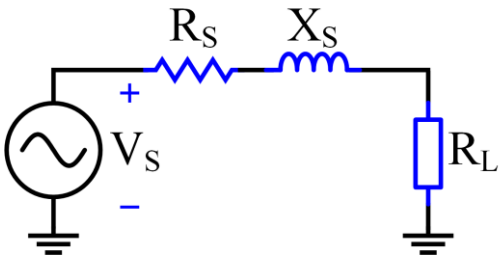
The positive sign means the sending end is generating reactive power.

Other answers using the exact power flow equations (not approximated) are also accepted.

b) Consider the characteristics of single-phase overhead lines suppling an impedance load, as shown in the figure below, and answer these questions.

i) The overhead line has an impedance of 0.5 + j4.0 Ω. The voltage at the sending end is 1300V. If a pure resistive load with impedance of 15.0 Ω is connected at the receiving end, calculate the terminal voltage magnitude at the receiving end, as well as the active power that consumed by the load. [5]

ii) For the line in i), consider a pure resistive load R_L connected at the receiving end. Calculate the maximum power that can be supplied to the load. (Hint: find the power delivered to R_L as a function of R_L and identify the maximum value of the function.) [5]



Solutions:

i)

As the line impedance is series with the load, the current through the transmission line and the load is

$$|I| = \frac{|V_S|}{|\widetilde{\boldsymbol{Z_S}} + \widetilde{\boldsymbol{Z_L}}|} = \frac{1300}{|0.5 + j4.0 + 15.0 + j0|} = 81.21\text{A}$$

The volage at the receiving end is

$$V_R = \frac{|\widetilde{\boldsymbol{Z_L}}|}{|\widetilde{\boldsymbol{Z_S}} + \widetilde{\boldsymbol{Z_L}}|} V_S = 1218.16\text{V}$$

The active power consumed by the load are then calculated as

$$P_L = R_L I_L^2 = 15.0 \times 81.21^2 = 98.93\text{kW}$$

ii)

Assume the load resistance is R_L, the current flow through the line and the load is expressed as

$$|I| = \frac{V_S}{|\widetilde{\boldsymbol{Z_S}} + \widetilde{\boldsymbol{Z_L}}|} = \frac{V_S}{\sqrt{(R_S + R_L)^2 + X_S^2}}$$

The active power consumed by the load is then expressed as

$$P_L = I^2 R_L = \frac{V_S^2 R_L}{(R_S + R_L)^2 + X_S^2} = \frac{V_S^2}{R_L + \frac{R_S^2 + X_S^2}{R_L} + 2R_S}$$

Therefore, the load power reaches its maximum when $R_L = \sqrt{R_S^2 + X_S^2} = 4.03\Omega$, and the maximum active power is

$$P_{Lmax} = \frac{V_S^2}{2\sqrt{R_S^2 + X_S^2} + 2R_S} = \frac{1300^2}{2 \times 4.03 + 2 \times 0.5} = 186.49\text{ kW}$$

4. This question addresses switch-mode power supplies. [30]

a) A Buck SMPS is to be designed to convert 80V to 25V with a maximum output power of 150W. It will operate in continuous mode. The following design choices have been made.

Switching frequency, $f = 20\text{kHz}$
Inductor, $L = 800\mu\text{H}$
Capacitor, $C = 1000\mu\text{F}$ with an effective series resistance, $R_{ESR} = 40\text{m}\Omega$

i) Calculate the duty cycle needed for the 80V to 25V voltage conversion ratio, neglecting the voltage drop on the MOSFET and diode. [4]

ii) Use the same input voltage and duty cycle as in a), find the minimum output current below which the Buch SMPS would enter discontinuous conduction mode. [4]

iii) Use the same input voltage and duty cycle as in a), and assume the circuit is operating at the maximum power point. Calculate the maximum instantaneous current that flows through the diode. [4]

iv) The MOSFET to be used will be chosen from a range that has the following properties when it is switched:

The turn-on energy loss of the MOSFET is 20.0 μJ
The turn-off energy loss of the MOSFET is 25.0 μJ

It is required that the junction temperature is 70°C or less in operation. The temperature of the ambient air is 40°C and the heatsink is such that the junction-to-air thermal resistance is R_{th}=15K/W.
Calculate the maximum value of the on-state resistance, R_{DS}, that the MOSFET can be allowed to have to meet this thermal design at the maximum output power of 150W. [8]

Solutions:

a)

i)

The duty cycle $\delta = \frac{V_O}{V_I} = \frac{25}{80} = 0.3125$

ii)

The minimum current equals half of the current ripple. The current ripple is calculated as

$$\Delta i_L = \frac{V_{in} - V_{out}}{L} t_{on} = \frac{V_{in} - V_{out}}{L} \frac{\delta}{f} = \frac{80 - 25}{800 \times 10^{-6}} \times \frac{0.3125}{20 \times 10^3} = 1.074\text{A}$$

Hence, the minimum current is $I_{min} = \frac{\Delta i_L}{2} = 0.537\text{A}$

iii)

The output current is firstly calculated with $I_o = \frac{P_o}{V_{out}} = \frac{150}{25} = 6\text{A}$

The current flow through the diode when the MOSFET is off. The diode reaches the maximum current at the switch-off moment, and it equals to the maximum current of the inductor, expressed as

$$I_{Dmax} = I_o + \Delta i_L = 6.537\text{A}$$

iv)

Firstly, calculate the maximum allowable power loss.

$$P_{loss}^{Max} = \frac{\theta_J - \theta_A}{R_{th}} = \frac{70 - 40}{15} = 2\text{W}$$

Then, calculate the switching power loss of the MOSFET

The switching loss of the MOSFET is:

$P_{sw} = f(E_{on} + E_{off}) = 20 \times 45 \times 10^{-6} = 0.9\text{W}$

The switch-off loss of the MOSFET is

$P_{off} = V_{in} I_{off}(1 - \delta) = 80 \times 300 \times 10^{-6} \times (1 - 0.3125) = 0.0165\text{W}$

The maximum conduction loss of the MOSFET is:

$$P_{cd}^{max} = P_{loss}^{max} - P_{sw} - P_{off} = 2 - 0.9 - 0.0165 = 1.0835\text{W}$$

For the conduction power loss, it can be expressed as

$$P_{cd} = I_o^2 R_{DS} \delta$$

The maximum value of the on-state resistance is then calculated with

$$R_{DS}^{max} = \frac{P_{cd}^{max}}{I_o^2 \delta} = \frac{1.0835}{6^2 \times 0.3125} = 0.0963\ \Omega$$

b) A second SMPS of Buck-Boost form is required to produce a -25 V output from the 80 V input. The circuit will be operated at a switching frequency of 30 kHz in discontinuous conduction mode. For operation at 150 W output power a duty cycle of 0.15 will be used.

i) Calculate the stored energy that should be present in the inductor at the end of the on-time to transfer 150 W under the stated conditions. [2]

ii) For the given input and output voltage settings, calculate the maximum on-time that can be used while remaining in the discontinuous mode. [3]

iii) For operation at 150 W output power a duty cycle of 0.15, calculate the inductor current at the end of the on-time. [3]

iv) Calculate the output voltage ripple across the capacitor if the effective series resistance of the capacitor is R_{ESR} = 15 mΩ (you may ignore the ripple across the capacitance of the capacitor). [2]

b)

i)

The energy stored in the inductor equals the energy taken away by the load over one cycle:

$E = P_{load} T = P_{load}/f_s$ = 150/30000 = 5 mJ

ii)

For critical cases, there is $\delta_{on} + \delta_{off} = 1$. For the buck-boost circuit, there satisfies

$$V_{in}\delta_{on} + V_{off}\delta_{off} = 0$$

Therefore, the critical δ_{on} is calculated as $\delta_{on} = \frac{V_{off}}{V_{off} - V_{in}} = \frac{-25}{-25-80} = 0.2381$

The corresponding switch-on time per cycle is $t_{on} = \frac{\delta_{on}}{f} = \frac{0.2381}{30000} = 7.937\ \mu\text{s}$

iii) Firstly, the inductance is calculated. At the end of the on-time, the maximum inductance current is expressed as $I_{max} = \frac{\delta V_{in}}{fL}$. The maximum power at the moment is calculated in i), which is expressed as $P_{max} = \frac{1}{2} L I_{max}^2 = \frac{\delta^2 V_{in}^2}{2f^2 L}$. Hence, the inductance is calculated as

$$L = \frac{\delta^2 V_{in}^2}{2f^2 P_{max}} = \frac{0.15^2 \times 80^2}{2 \times 30000^2 \times 5 \times 10^{-3}} = 16\mu\text{H}$$

Therefore, the maximum current at the end of the on-time is

$$I_{max} = \frac{\delta V_{in}}{fL} = \frac{0.15 \times 80}{30000 \times 16 \times 10^{-6}} = 25\text{A}$$

iv) The peak-to-peak current ripple is equal to the maximum inductance current, expressed as

$\Delta I = I_{max} = 25\text{A}$

Then, the voltage ripple is calculated as

$\Delta V = \Delta I \, R_{ESR} = 25 \times 15 \times 10^{-3} = 0.375\text{V}$